The Physics Behind a Rocket Launch Simulator

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1 Introduction

This report aims to explain the all the physics and mathematics behind the Python launch simulator and also describes the main idea of the simulation's program code.

2 The Physics Behind a Rocket Flight

In order to determine the flight parameters of the rocket as a function of time we need to write the equations that describe the motion of the rocket.To do this we start from the law of change in the mechanical momentum of a system of objects(the rocket itself and its exhaust gases) : $\frac{d\vec{p}}{dt} = \sum \vec{F_e}$ Where $\frac{d\vec{p}}{dt}$ is the first derivative over time of the system's momentum (the vector sum of the momentum of all objects in the system) and $\sum \vec{F_e}$ is the vector sum of all external forces action on the system. In this simulation the buoyant force and the lift force form the pressure gradient are neglected and for simplicity we assume that o the rocket are acting only the gravitational force, the thrust force and the force of air resistance. When we write the equation from the law of change change in the momentum: $d\vec{p} = (m(t)\vec{g} + \vec{f})dt = \vec{p}(t+dt) - \vec{p}(t)$ The force of air resistance: $\vec{f} = \frac{1}{2}C\rho.Sv^2\vec{e_r}$ Where C is the air resistance coefficient and it is function of the angle of attack θ (in our case $\theta = 0$-the velocity vector is collinear with the rocket's axis) and the mach number $M = \frac{v_r}{c}$ - the ratio of the rocket's velocity and the speed of sound. In the simulation different coefficients are used for the different mach number in order to simulate this dependency. The air density is a function of the altitude y: Where C Where ρ_0 is the air density at sea level and H is a reference altitude of 10.4 km. $S = \pi.r^2$ is the area of the cross-section of the rocket.

v is the rocket's velocity relative to the air and when wind with speed v_w is blowing toward the rocket for the vector summation of the velocities we get that: $\vec{v} = \vec{v_r} - \vec{v_w}$ where $\vec{v_r}$ is the rocket's velocity relative to the ground. For the components of $\vec{v}$ we get: $v_x = v_rx + v_w,\ v_y = v_ry$

($v_w < 0$ because the wind is blowing in the opposite direction of the x-axis). The $\vec{e_r} = -\frac{\vec{v}}{|\vec{v}|}$ vectorizes (gives direction) the air resistance. For the change in system's momentum we get:

$$\vec{p}(t+dt) - \vec{p}(t) = (-\frac{1}{2}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}(\vec{v}) + m\vec{g})dt$$

The first derivative of the mass of the rocket over time $\frac{dm}{dt} = \mu < 0$ (mass is decreasing over time) $dm = \mu.dt < 0 => |dm| = -dm$
When particle with infinitesimal mass exits the rocket:

$$\vec{p}(t) = m(t)\vec{v_r}(t)$$
$$\vec{p}(t+dt) = (m(t)+dm)(\vec{v_r}(t)+d\vec{v}) - dm(\vec{v_r}(t)+d\vec{v}+\vec{u})$$
$$=> md\vec{v} - dm\vec{u} = (-\frac{1}{2}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}(\vec{v}) + m\vec{g})dt$$
$$\frac{d\vec{v}}{dt} - \frac{dm}{dt}\frac{1}{m}\vec{u} = -\frac{1}{2m}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}(\vec{v}) + \vec{g}$$
$$\frac{d\vec{v}}{dt} = -\frac{1}{2m}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}(\vec{v}) + \vec{g} + \frac{dm}{dt}\frac{1}{m}\vec{u}$$

Where $\vec{u} = -|\vec{u}|\frac{\vec{v_r}}{|\vec{v_r}|}$ is the velocity of the exhaust gases relative to the rocket. $|\vec{u}| = Isp.g$ where Isp is called specific impulse and measures the efficiency of the rocket engine.

$$\frac{d\vec{v}}{dt} = \begin{bmatrix} \frac{dv_rx}{dt} \\ \frac{dv_ry}{dt} \end{bmatrix} = -\frac{1}{2m}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}\begin{bmatrix} v_rx + v_w \\ v_ry \end{bmatrix} + \begin{bmatrix} 0 \\ -g \end{bmatrix} - \frac{dm}{dt}\frac{1}{m}\begin{bmatrix} v_rx \\ v_ry \end{bmatrix}\frac{Isp.g}{\sqrt{(v_rx^2 + v_ry^2)}}$$

From here we get the equations for x and y component of the acceleration of the rocket:

$$a_x = \frac{dv_rx}{dt} = -\frac{1}{2m}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}(v_rx + v_w) - -\frac{dm}{dt}\frac{1}{m}v_rx\frac{Isp.g}{\sqrt{(v_rx^2 + v_ry^2)}}$$
$$a_y = \frac{dv_ry}{dt} = -g - \frac{1}{2m}CS\rho.\sqrt{(v_rx + v_w)^2 + v_ry^2}v_ry - -\frac{dm}{dt}\frac{1}{m}v_ry\frac{Isp.g}{\sqrt{(v_rx^2 + v_ry^2)}}$$

3 Simulation

In the program we firstly import all necessary libraries for the solution of the differential equations. We create vector that contains all parameters (the coordinates of the rocket, the components of the rocket's velocity and the mass).

The components of the derivative of this vector over time are the components of the velocity, acceleration, and mass changing rate. For the components of the vector's derivative we write our differential equations. The value of mass changing rate is controlled by 'if' statement and is set to be constant when the engine is running and zero when the engine is not running. The values of the air resistance coefficient is also controlled via 'if' statement for the different mach numbers. The differential equations are solved via solve-ivp solver (using Runge-Kuta method for numerical solving of differential equations) from scipy.integrate and event is used to stop the solving when the rocket hits the ground. The trajectory and the rocket's velocity graphs are visualized via pyplot from matplotlib and the graphs are animated using 'update' function and FuncAnimation from matplotlib.animation. The initial launch angle in degrees and initial velocity are taken from the console.